Half-Wormholes as Wiener Chaos: Correlated-Replica Spectroscopy, Exact Pfaffian Sectors, and Boundary Moment Indeterminacy

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Abstract

The fixed-coupling partition function of one-time-point SYK is a homogeneous multilinear polynomial of degree $n = N/q$ in independent Gaussian couplings. Its square therefore has a canonical Wiener–Itô chaos decomposition. This is a probabilistic refinement of the exact half-wormhole expansion already constructed by Mukhametzhanov, not a replacement of that prior result. We derive five consequences of the canonical grading. First, two replicas with coupling correlation ρ obey $\mathbb{E}[z(J)z(J')] = \rho^n \mathbb{E}[z^2]$; scanning ρ diagonalises and measures the chaos-sector norms of z^2. Second, for $q = 2$ we reduce the previously known general variance sum to the closed formula $\mathbb{E}[W_{2k}^2] = (2n)!\binom{2(n-k)}{n-k}4^{-(n-k)}$. It yields an explicit correlated determinant formula for two Gaussian antisymmetric matrices. Third, the same profile gives $\mathbb{E}[z^4]/\mathbb{E}[z^2]^2 = 2n+1$ and quantifies the failure of a wormhole-plus-Wick-square truncation at $q = 2$. Fourth, Gaussian hypercontractivity supplies non-asymptotic, single-draw reconstruction bounds sector by sector. Fifth, the boundary moment problem for the law of z is determinate for $n \leq 2$ and indeterminate for every $n \geq 3$. We prove the latter by a conditional-Gaussian density lower bound and Krein's criterion. The result is a complete statement for the zero-dimensional model. It does not establish a dynamical bulk geometry or a complete theory of quantum gravity.

1 Introduction

In the one-time-point SYK model, fixed-coupling factorisation is represented by a finite polynomial identity. Saad, Shenker, Stanford and Yao identified the wormhole and coupling-sensitive half-wormhole saddles [13]. Mukhametzhanov subsequently constructed an exact finite expansion of z^2, including orthogonality, closed sector variances and the normal-ordered-square interpretation of the leading half-wormhole [9]. Related graded and Hermite structures appear in Tian and Yang [18]. Those results are prior art and are used here as such. After matching conventions, Mukhametzhanov's term Φ_{n-k} is precisely the chaos projection W_{2k} below. Thus neither the finite sector decomposition nor its general variance sum is claimed as new; the $q = 2$ binomial form is a closed reduction supplied with an independent generating-function proof.

Our starting point is narrower: over the Gaussian coupling ensemble, z^2 has a unique decomposition into eigenspaces of the Ornstein–Uhlenbeck (OU) semigroup. Identifying this canonical grading makes the correlated-replica kernel, Gaussian hypercontractivity and the Hamburger moment problem available without introducing new saddles. A targeted search recorded in the July 2026 source project found no substantive duplicate of the combined results in the half-wormhole, random-matrix and moment-problem literature. That historical search is bounded and has not been replaced by a comprehensive current novelty audit for this repository edition.

The paper develops the following exact statements and applications:

1. the ρ^n two-replica identity and OU sector spectroscopy;

2. a closed $q = 2$ reduction of the known sector-variance sum, an independent enumeration, and the correlated antisymmetric-determinant corollary;

3. an explicit fourth-moment and truncation diagnosis, together with single-draw bounds;

4. the sharp determinacy dichotomy at chaos degree $n = 2/3$ and its spectral consequences.

We separate exact finite-N theorems from asymptotic interpretations. In particular, the standard Nualart–Peccati fourth-moment equivalence applies directly to sequences in a *fixed* Wiener chaos [11, 12]. In the conventional SYK scaling with fixed q and $N \to \infty$, the chaos order $n = N/q$ grows; fourth moments alone are then not advertised here as a central-limit proof. The L^2 truncation statements we use instead follow from the exact sector variances.

2 Model and canonical grading

Let $q \geq 2$ be even, $N = qn$, and let A run over the q-subsets of $[N]$. The couplings J_A are independent standard normal variables. Berezin integration gives, up to an irrelevant global phase,

$$z(J) = \sum_{P \in \mathrm{Part}_q(N)} \epsilon(P) \prod_{A \in P} J_A, \tag{1}$$

where P partitions $[N]$ into n blocks of size q and $\epsilon(P) = \pm 1$ is the Grassmann sign. For $q = 2$, $z(J) = \mathrm{Pf}\, J$ for the antisymmetric matrix with upper-triangular entries J_{ij}.

Let $L^2(\gamma) = \bigoplus_{d \geq 0} \mathcal{C}_d$ be the Wiener–Itô decomposition of the Gaussian coupling space [3, 4]. The OU operator is

$$(T_\rho F)(J) = \mathbb{E}_K F\Big(\rho J + \sqrt{1-\rho^2} K\Big), \qquad -1 \leq \rho \leq 1, \tag{2}$$

and acts on $\mathcal{C}_d$ by multiplication by ρ^d.

Proposition 2.1 (Purity and symmetry)**.** *$z \in \mathcal{C}_n$, $\mathbb{E}[z^2] = |\mathrm{Part}_q(N)|$, and the law of z is symmetric.*

Proof. Every monomial in (1) is a product of n distinct first Hermite polynomials, hence belongs to $\mathcal{C}_n$. Distinct partitions give distinct monomials, so orthogonality gives the second statement. Fix a vertex, say 1, and change the sign of every coupling J_A with $1 \in A$. Every partition contains exactly one such block, so $z \mapsto -z$, whereas the joint Gaussian law is unchanged. Thus all odd moments vanish and the law is symmetric. □

Theorem 2.2 (Correlated-replica identity)**.** *Let $J' = \rho J + \sqrt{1-\rho^2} K$ with K an independent coupling draw. Then*

$$\mathbb{E}[z(J) z(J')] = \rho^n \mathbb{E}[z^2]. \tag{3}$$

More generally, if $X = \sum_d X_d$ and $Y = \sum_d Y_d$ are their chaos decompositions, then

$$\mathbb{E}[X(J) Y(J')] = \sum_d \rho^d \mathbb{E}[X_d Y_d]. \tag{4}$$

Proof. Equation (4) is Mehler's identity $\mathbb{E}[X(J)Y(J')] = \langle X, T_\rho Y \rangle$. Proposition 2.1 then gives (3). □

Define the canonical half-wormhole sectors by

$$z^2 = \sum_{k=0}^{n} W_{2k}, \qquad W_{2k} \in \mathcal{C}_{2k}, \qquad \mathbb{E}[W_{2k} W_{2\ell}] = 0 \quad (k \neq \ell). \tag{5}$$

Thus $W_0 = \mathbb{E}[z^2]$, while $W_{2n} =: z^2 :$ is the Wick square. This decomposition is canonical because the $\mathcal{C}_d$ are the distinct OU eigenspaces. It need not coincide term-by-term with an arbitrary orthogonal expansion once $n \geq 3$. For the normal-ordered expansion of Mukhametzhanov [9], however, homogeneity gives the exact identification $\Phi_{n-k} = W_{2k}$ after matching the coupling normalisation.

Applying (4) to z^2 gives

$$\mathbb{E}[z^2(J)z^2(J')] = \sum_{k=0}^{n} \rho^{2k} \mathbb{E}[W_{2k}^2]. \tag{6}$$

Hence correlation in coupling space is a sector spectrometer: the coefficients of the polynomial in ρ^2 are precisely the sector norms.

3 A low-order sector at even block size

The general sector variances are prior results of Mukhametzhanov [9]. The following elementary specialization records explicitly how even block size enters completion-sign coherence; it is supporting finite-model algebra rather than a new general variance theorem.

Proposition 3.1 (The fourth-chaos sector at even q)**.** *Let q be even and $n = N/q \geq 2$. Write $M_r = |\operatorname{Part}_q(N-2q)|$ and $M_2 = |\operatorname{Part}_q(2q)|$, with the empty partition counted once. Then*

$$\mathbb{E}[W_4^2] = 4M_r^2 \left[\frac{1}{2} \binom{N}{q} \binom{N-q}{q} + \binom{N}{2q} \binom{M_2}{2} \right]. \tag{7}$$

Proof. In the Hermite expansion of the square of (1), a fourth-chaos monomial has one of two forms. First, $H_2(J_A)H_2(J_B)$ occurs for disjoint blocks A, B. Its coefficient is M_r, because every completion of the other vertices contributes with squared partition sign one. There are $\frac{1}{2}\binom{N}{q}\binom{N-q}{q}$ such unordered pairs, and each monomial has squared Gaussian norm $2 \cdot 2 = 4$.

Second, a product of four distinct first-order variables comes from two distinct splittings of the same $2q$-vertex support into q-blocks. The relative partition sign is independent of the common completion: moving one entire q-block past another contributes $(-1)^{q^2} = 1$. The two orders of the pair of splittings therefore give coefficient $\pm 2M_r$. There are $\binom{N}{2q}\binom{M_2}{2}$ distinct monomials of this kind, each of norm one. Distinct splittings cannot share exactly one block, since their complementary block would then also agree. For the same reason there is no additional $H_2(J_A)J_BJ_C$ type: after the shared blocks are removed, two partitions differing by one block must use the same complement. Hermite orthogonality now gives (7). □

Remark 3.2 (A sign-scope diagnostic). For odd q, sorting blocks lexicographically and then taking the permutation sign defines a formal Gaussian partition polynomial, but the completion signs need not cohere. At $(N, q) = (12, 3)$, exact completion-sign enumeration gives $\mathbb{E}[W_4^2] =$ 17,580,480, whereas blindly using (7) gives 20,328,000. This is a diagnostic of the even-q hypothesis, not a claim about an odd-q Grassmann SYK partition function.

4 Exact Pfaffian sector spectrum

For $q = 2$, ordered pairs of perfect matchings provide a closed enumeration.

Theorem 4.1 ($q = 2$ sector profile)**.** *For $z = \operatorname{Pf} J$ with J a $2n \times 2n$ Gaussian antisymmetric matrix,*

$$\boxed{\mathbb{E}[W_{2k}^2] = (2n)! \binom{2m}{m} 4^{-m}, \qquad m = n-k, \quad 0 \leq k \leq n.} \tag{8}$$

Equivalently,

$$\sum_{n\geq 0}\sum_{k=0}^{n}\mathbb{E}[W_{2k}^2]x^{2k}\frac{y^{2n}}{(2n)!}=\frac{1}{(1-x^2y^2)\sqrt{1-y^2}}. \tag{9}$$

Proof. Expand z^2 over ordered perfect matchings (P,Q). Their symmetric difference is a vertex-disjoint union U of alternating even cycles of length at least four; shared edges contribute $J_e^2=H_2(J_e)+1$. A chaos monomial is therefore labelled by a set T of excited shared edges and by U, with order $2|T|+|E(U)|$. Each alternating cycle has two splittings into (P,Q), and the unused vertices carry an arbitrary common matching. Upon squaring a coefficient, the two common matchings form another union of doubled edges and alternating cycles. Sign coherence holds because matching edges have even block size.

The squared norm is thus the exponential generating function of labelled components of four types: an excited doubled edge of weight $2x^2$, a first-family cycle of length $2j\geq 4$ of weight $4x^{2j}$, an unexcited doubled edge of weight 1, and a second-family cycle of length $2j\geq 4$ of weight 2. Since the number of undirected alternating cycles on $2j$ labelled vertices supplies the standard $1/(2j)$ exponential weight, the exponential formula gives

$$\exp\left[x^2y^2+\sum_{j\geq 2}\frac{(xy)^{2j}}{j}+\frac{y^2}{2}+\sum_{j\geq 2}\frac{y^{2j}}{2j}\right]=\frac{1}{(1-x^2y^2)\sqrt{1-y^2}}.$$

Extracting $x^{2k}y^{2n}$ and using $[u^m](1-u)^{-1/2}=\binom{2m}{m}4^{-m}$ proves (8). □

Equation (8) is a closed reformulation of the $q=2$ case of Mukhametzhanov [9, Eq. (47)]. With unit coupling variance and $m=n-k$, that formula reads

$$\mathbb{E}[W_{2k}^2]=\frac{(2n)!}{4^n}\binom{2m}{m}\sum_{a=0}^{k}\binom{2a}{a}\binom{2(k-a)}{k-a}.$$

The convolution is 4^k, since the square of $(1-u)^{-1/2}$ is $(1-u)^{-1}$. Thus the compact profile is not a new general sector-variance theorem. The proof above provides an independent labelled-component derivation and a convenient generating-function form for the correlated-determinant application below.

Two checks are immediate. At $k=0$, (8) equals $\mathbb{E}[z^2]^2=((2n-1)!!)^2$. Summing over k and using

$$\sum_{m=0}^{n}\binom{2m}{m}4^{-m}=(2n+1)\binom{2n}{n}4^{-n} \tag{10}$$

gives

$$\frac{\mathbb{E}[z^4]}{\mathbb{E}[z^2]^2}=2n+1. \tag{11}$$

Corollary 4.2 (Correlated antisymmetric determinants)**.** *If $J'=\rho J+\sqrt{1-\rho^2}K$ with J,K independent Gaussian antisymmetric $2n\times 2n$ matrices, then*

$$\mathbb{E}[\det J\,\det J']=(2n)!\sum_{m=0}^{n}\binom{2m}{m}4^{-m}\rho^{2(n-m)}. \tag{12}$$

Proof. $\det J=\mathrm{Pf}(J)^2=z^2$. Combine (6) with Theorem 4.1. □

Lemma 4.3 (Gaussian Pfaffian convolution)**.** *Let M be a fixed real antisymmetric $2n \times 2n$ matrix, let the upper entries of K be independent standard Gaussian variables, and let $c \in \mathbb{R}$. For a principal submatrix M_S use $\det M_\varnothing = 1$ and $(-1)!! = 1$. Then*

$$\mathbb{E}_K \det(M + cK) = \sum_{r=0}^{n} c^{2r}(2r-1)!! \sum_{\substack{S\subseteq[2n] \\ |S|=2n-2r}} \det M_S. \tag{13}$$

Proof. The Pfaffian minor expansion is

$$\operatorname{Pf}(M + cK) = \sum_{\substack{S\subseteq[2n] \\ |S| \text{ even}}} \epsilon(S) c^{|S^c|/2} \operatorname{Pf}(M_S) \operatorname{Pf}(K_{S^c}),$$

where $\epsilon(S)$ is the shuffle sign. Gaussian Pfaffian polynomials on distinct vertex sets are orthogonal: their matching monomials use different edge sets, so every cross expectation contains a Gaussian variable to an odd power. For $|S^c| = 2r$, the diagonal norm is the number of perfect matchings, $\mathbb{E}[\operatorname{Pf}(K_{S^c})^2] = (2r-1)!!$. Squaring the expansion, taking expectation and using $\operatorname{Pf}(M_S)^2 = \det M_S$ gives (13). □

This elementary minor identity is included as a supporting Gaussian calculation and an independent way to check Pfaffian moment manipulations; no separate priority claim is assigned to it.

Formula (12) is distinct from the general characteristic-polynomial averages treated in random-matrix theory [2, 8]. We have not found this correlated antisymmetric specialisation stated in the searched literature. This negative search is recorded as evidence, not as a mathematical proof of priority.

5 Truncation and single-draw control

At $q = 2$, the fraction of the total L^2 norm in sector k is proportional to $\binom{2(n-k)}{n-k} 4^{-(n-k)}$. The Wick-square fraction is

$$\frac{\mathbb{E}[W_{2n}^2]}{\mathbb{E}[z^4]} = \frac{1}{(2n+1)\binom{2n}{n}4^{-n}} \sim \frac{\sqrt{\pi}}{2\sqrt{n}}. \tag{14}$$

Thus no top sector dominates, and the wormhole-plus-Wick-square truncation fails in relative L^2 as $n \to \infty$. Equation (11) also shows divergence of the normalised fourth moment. This rules out Gaussian convergence in fourth moment; we do not infer a general weak-limit theorem from this fact alone.

For any sector, hypercontractivity converts its exact norm into a single-draw bound.

Lemma 5.1 (Chaos tail envelope)**.** *If $X \in \mathcal{C}_d$, then for $t > 0$*

$$\mathbb{P}(|X| \geq t\|X\|_2) \leq B_d(t) := \inf_{p\geq 2} (p-1)^{pd/2} t^{-p}. \tag{15}$$

In particular $B_d(t) \leq t^{-2}$, and for $t \geq (2e)^{d/2}$, $B_d(t) \leq \exp[-(d/2e)t^{2/d}]$.

Proof. Gaussian hypercontractivity gives $\|X\|_p \leq (p-1)^{d/2}\|X\|_2$ [4, 10]. Markov's inequality gives (15). The two displayed bounds follow from $p = 2$ and from the admissible choice $p = t^{2/d}/e$, respectively. □

Put $f_k = \mathbb{E}[W_{2k}^2]/\mathbb{E}[z^2]^2$ and $E_{\mathrm{tr}} = z^2 - \mathbb{E}[z^2] - W_{2n} = \sum_{k=1}^{n-1} W_{2k}$.

Theorem 5.2 (Single-draw reconstruction)**.** *For $n \geq 2$ and any $\varepsilon_k > 0$ with $\sum_{k=1}^{n-1} \varepsilon_k = \varepsilon$,*

$$\mathbb{P}\big(|E_{\mathrm{tr}}| \geq \varepsilon \mathbb{E}[z^2]\big) \leq \min\left\{ \frac{\sum_{k=1}^{n-1} f_k}{\varepsilon^2}, \sum_{k=1}^{n-1} B_{2k}\left(\frac{\varepsilon_k}{\sqrt{f_k}}\right) \right\}. \tag{16}$$

Proof. Orthogonality gives $\mathbb{E}[E_{\mathrm{tr}}^2]/\mathbb{E}[z^2]^2 = \sum_{k=1}^{n-1} f_k$, hence the first bound. For the second, if $|\sum W_{2k}| \geq \sum \varepsilon_k \mathbb{E}[z^2]$, at least one sector exceeds its threshold. Apply Lemma 5.1 and the union bound. □

Theorem 5.2 is non-asymptotic; its probability is over one Gaussian coupling draw. It is not a deterministic error guarantee for every prescribed realization. For $n = 1$, the intermediate-sector sum is empty and $E_{\mathrm{tr}} = 0$ identically. For this case the probability of a positive error threshold is zero. The dimension-free constants are not asserted optimal. When prior exact sector estimates show $\sum_{k=1}^{n-1} f_k \to 0$ in a specified large-N regime [9], the first term already proves relative L^2 reconstruction; the tail term refines atypical events sector by sector.

6 Boundary moment determinacy

Let μ be the law of z. In the baby-universe/GNS dictionary, the moments $\mu_j = \mathbb{E}[z^j]$ are multi-boundary data, and multiplication by z acts on the closure of polynomials in $L^2(\mu)$ [5, 7]. Whether all moments determine μ is exactly the Hamburger moment problem.

Theorem 6.1 (Determinacy dichotomy)**.** *For the one-time-point model (1):*

1. *if $n \leq 2$, the law μ is Hamburger determinate;*
2. *if $n \geq 3$, the law μ is Hamburger indeterminate.*

Proof. For $n \leq 2$, hypercontractivity applied to $z \in \mathcal{C}_n$ gives

$$\mu_{2m}^{1/(2m)} = \|z\|_{2m} \leq (2m-1)^{n/2} \|z\|_2.$$

Therefore $\sum_m \mu_{2m}^{-1/(2m)}$ dominates a constant multiple of $\sum_m m^{-n/2}$ and diverges. Carleman's criterion implies determinacy [1, 14].

Now let $n \geq 3$. Expand along blocks containing vertex 1:

$$z = \sum_{A \ni 1} \epsilon_A J_A C_A, \tag{17}$$

where every cofactor C_A uses only couplings on subsets not containing 1 and is independent of the family $\{J_A : A \ni 1\}$. Conditionally on the cofactors, $z \sim \mathcal{N}(0, S^2)$ with $S^2 = \sum_{A \ni 1} C_A^2$. Since the cofactor vector is not identically zero and its common zero set is algebraic of Gaussian measure zero, $S > 0$ almost surely. Thus μ has a strictly positive Gaussian-mixture density

$$f(t) = \mathbb{E}\left[(2\pi S^2)^{-1/2} \exp\left(-\frac{t^2}{2S^2}\right) \right]. \tag{18}$$

We need a lower tail bound on f. Fix a partition $\{A_0, B_2, \ldots, B_n\}$ with $1 \in A_0$. On the event that $J_{B_j} \in [a, a+1]$ for $j = 2, \ldots, n$ and all other couplings not containing 1 have absolute value at most one, the leading monomial of C_{A_0} has degree $n-1$. All its other monomials, and every C_A with $A \neq A_0$, contain at most $n-2$ of the selected large variables. Hence, for a above a model-dependent constant,

$$c a^{n-1} \leq S \leq C a^{n-1}, \qquad \mathbb{P}(E_a) \geq c_1 e^{-c_2 a^2}.$$

Insert this event into (18) and choose $a = t^{1/n}$. For sufficiently large t,

$$f(t) \geq c_3 t^{-(n-1)/n} \exp[-c_4 t^{2/n}]. \tag{19}$$

By symmetry, the same lower bound holds at negative infinity. Because $2/n < 1$, (19) makes the positive part of $-\log f(t)/(1+t^2)$ integrable outside a compact interval. For completeness, choose $0 < s_- < s_+ < \infty$ with $\mathbb{P}(s_- \leq S \leq s_+) > 0$. Restricting the mixture to this event bounds f uniformly away from zero on every compact interval. Also $(\log f)_+ \leq f$ where $f \geq 1$, so its integral is finite. Thus the compact part of the logarithmic integral is well defined and finite. Together these estimates give

$$\int_{\mathbb{R}} \frac{-\log f(t)}{1+t^2}\, dt < \infty.$$

Krein's criterion therefore implies Hamburger indeterminacy [6, 16]. This proves the dichotomy. □

The odd-degree case $n \geq 3$ is already covered by prior results on polynomial forms in normally distributed variables [15]. We include the direct density proof because it treats this hyperpfaffian family uniformly, including even degree. Slud also studies even-degree cases, and products of normal variables have a related determinacy threshold [17]. A full theorem-level comparison with those results and subsequent literature remains necessary before assigning originality to the even-degree hyperpfaffian application. We therefore claim a self-contained model-specific proof here, rather than a new general theorem about moment problems for Gaussian polynomials.

For $n \geq 3$, the symmetric Jacobi operator associated with the boundary moments is not essentially self-adjoint on polynomials. Its self-adjoint extensions within the moment Hilbert space give the one-parameter N-extremal subfamily of the full Nevanlinna parametrization; these measures are discrete [1, 14]. The physical coupling law (18) is continuous and non-extremal, so polynomials are not dense in its full L^2 space. This does not make the GNS construction inconsistent. It says only that one-time boundary moments do not select the microscopic coupling measure uniquely once one boundary amplitude is a Gaussian polynomial of degree at least three.

6.1 A finite quadrature calibration

Finite moment data also give a useful, classical compression of the boundary operator. The $m \times m$ Jacobi compression on polynomials of degree less than m yields the m-node Gaussian quadrature rule, exact through degree $2m-1$ [1, 14]. These inputs are ordinary moments, not connected cumulants. Finite exactness alone does not select a measure in an indeterminate moment problem; limits there require an additional extension or boundary prescription.

For an explicit determinate example take $(N, q) = (4, 2)$. The Pfaffian is $z = u \cdot v$ for independent standard Gaussian vectors $u, v \in \mathbb{R}^3$, after one harmless coupling sign change. Conditioning on u gives

$$\mu_{2j} = (2j-1)!!(2j+1)!!, \qquad \mu_{2j+1} = 0.$$

In particular $(\mu_0, \mu_2, \mu_4, \mu_6) = (1, 3, 45, 1575)$. The monic degree-four orthogonal polynomial is

$$p_4(x) = x^4 - 40x^2 + 75.$$

Indeed $\int p_4\,d\mu = 45 - 40\cdot 3 + 75 = 0$ and $\int x^2 p_4\,d\mu = 1575 - 40\cdot 45 + 75\cdot 3 = 0$, while the odd orthogonality equations follow by symmetry. The four nodes and their individual weights are

$$x = \pm\sqrt{20 - 5\sqrt{13}}, \qquad w = \frac{1}{4} + \frac{17}{20\sqrt{13}},$$
$$x = \pm\sqrt{20 + 5\sqrt{13}}, \qquad w = \frac{1}{4} - \frac{17}{20\sqrt{13}}.$$

The weights are positive and the resulting rule reproduces all moments through order seven. This is an exact finite calibration of classical Gauss quadrature, not evidence for infinite-order Airy indeterminacy or for a uniquely selected completion in the $n \geq 3$ case.

7 Scope and conclusion

The exact half-wormhole expansion, its identification with the chaos projections, and its general sector-variance sum are not new. The paper develops the correlated-replica use of the OU grading and the additional model-specific consequences presented here; originality of those applications remains subject to the comparisons stated above. Equations (3) and (6) turn correlated replicas into a sector probe. Theorem 4.1 gives a closed Pfaffian reduction and independent proof, while Corollary 4.2 gives the correlated-determinant consequence. Theorem 5.2 is a finite single-draw statement. Theorem 6.1 sharply separates determinate and indeterminate boundary laws.

The limitations are equally sharp. The model is zero-dimensional; "wormhole" labels the ensemble-connected term, not a derived spacetime geometry. Gaussianity is essential to the OU and hypercontractive form used here. A thermal partition function has an infinite chaos expansion. Finally, when the chaos order grows with N, the fixed-chaos fourth-moment theorem cannot by itself prove a central limit. These boundaries prevent the manuscript from being read as a construction of complete quantum gravity; it is a rigorous result about a solvable factorisation model.

Reproduction scope. The portable revision check verifies the central-binomial convolution and its reduction of the prior variance formula for $1 \leq n \leq 12$ and all $0 \leq k \leq n$, the corresponding fourth-moment sums, and the empty intermediate-sector case $n = 1$. Additional exact checks cover the displayed even/odd sign diagnostic, the four-node quadrature calibration, and direct Gaussian-Pfaffian convolution at $n = 1, 2, 3$. These finite exact checks corroborate the displayed formulas; the analytic proofs establish the all-size and density statements. Historical supplementary code is retained unchanged and its Monte Carlo loops were not rerun.

AI-assisted workflow and unresolved attribution

The repository operator states:

> I genuinely do not know how authorship, intellectual credit, or ownership should be attributed for these results.

The reported workflow was to ask AI systems to survey the literature, autonomously select open problems, and attempt to solve them. The operator reports that the earlier work was led mainly by Fable 5, whereas results from July 2026 onward were predominantly produced by GPT models. DeepSeek and multiple GPT models also participated in cross-review, checking, and manuscript preparation and organization.

This is the operator's reported chronological history, not a complete audited allocation of individual contributions. An earlier result is not reattributed merely because it was revised or packaged after July. Human author names and affiliations are intentionally blank. These

results have not undergone human expert review; automated checks and AI reviews are further AI review, not human peer review. Historical source dates, present packaging dates, and demonstrable public-disclosure dates are recorded separately. No authorship, ownership, journal acceptance, or publication-priority determination is made here.

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